

Amartya Paul, Senior Research Fellow National Institute of Technology Meghalaya

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Professional Summary

- A researcher specializing in microwave engineering, electromagnetic sensing, and AI-assisted RF system optimization, with a strong track record in developing advanced passive filters, antennas, and power amplifier architectures. My work spans experimental prototyping, food quality analysis through RF, and 3D microwave imaging for non-destructive evaluation of civil and structural materials. I have extensive experience with CST, HFSS, ADS, and VNA-based measurement protocols. Motivated to advance next-generation reconfigurable EM systems, I aim to contribute to AI-empowered antenna design and adaptive RF sensing technologies for 6G and beyond.

Education

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| Ph.D. (Pursuing) | ■ Electronics and Communication Engineering , National Institute of Technology Meghalaya (2022 – 2026)
Thesis title: <i>ML-Assisted Real-Time System Development for Identification and Quantification of Materials</i> . |
| M.Tech. | ■ Electronic and Communication Engineering North Eastern Hill University (2020 – 2022)
Thesis title: <i>Compact Multiband Bandpass Filters Using Substrate Integrated Waveguide with Fractal Resonators</i> (CGPA: 8.97). |
| B.E. | ■ Electronics and Communication Engineering SLIET, Punjab (2017 – 2020)
Project title: <i>Portable Health Checker Device with Blood Oxygen Monitor</i> (CGPA: 7.92). |
| Diploma | ■ Electronics and Communication Engineering SLIET, Punjab (2014 – 2017)
(CGPA: 8.27). |

Experience

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| Senior Research Fellow | ■ National Institute of Technology Meghalaya (September 2024 – Present) : Working on advanced microwave sensing systems for material characterization, structural health monitoring, and AI-driven RF modelling. Responsibilities: i) Designed, fabricated, and experimentally validated multiple microwave filters and sensing structures, including log-periodic feedline-based ring resonators and dual-band architectures. ii) Conducted extensive VNA-based measurements under controlled temperature and environmental conditions with strict experimental protocols. iii) Developed 3D microwave imaging frameworks for analyzing bitumen flow values, porosity, and mechanical behaviour using angle-resolved and height-resolved measurements. iv) Implemented AI/ML models to optimize EM structures, including ensemble learning, surrogate modelling, and parameter-level regression for sensitivity enhancement. v) Performed thermal analysis and reliability estimation of GaN power amplifiers using predicted/ measured junction temperatures. vi) Published research outcomes in IEEE and Elsevier journals and reviewed manuscripts for IEEE Sensors letters journals. |
| Junior Research Fellow | ■ National Institute of Technology Meghalaya (August 2022 – August 2024) : Worked on microwave non-destructive evaluation systems and electromagnetic material characterization. Responsibilities: i) Designed hollow circular microstrip ring filters and conducted experiments with cylindrical bitumen samples measured across height-wise and angular intervals. ii) Performed dielectric constant estimation of VOCs and studied sensor sensitivity to minute permittivity variations. iii) Built multiple datasets from repeated measurements and applied statistical analysis to evaluate variability across replications, experimenters, and operating conditions. iv) Ensured strict adherence to experimental consistency, calibration, and error minimization procedures. |

Research Publications

Journal Articles

- 1 A. Paul, G. Bhargava, P. K. Gautam, and S. Majumdar, "Non-destructive evaluation of strength and porosity of bituminous mixes using log-periodic feedlines-based ring filter," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–10, 2024. [DOI: 10.1109/TIM.2024.3381667](#).
- 2 A. Paul, G. Bhargava, D. Adak, S. Dutta, and S. Majumdar, "Design and validation of a microstrip log-periodic feedline based filter for microwave imaging of rebar," *IEEE Transactions on Instrumentation and Measurement*, 2024. [DOI: 10.1109/TIM.2024.3481566](#).
- 3 A. Paul, H. Kumari, and S. Majumdar, "Dual-band microwave spectroscopy and machine learning for rapid milk adulteration detection," *IEEE Transactions on AgriFood Electronics*, pp. 1–8, 2025. [DOI: 10.1109/TAFE.2025.3614005](#).
- 4 A. Paul, H. Kumari, S. Das, and S. Majumdar, "Low cost microwave system development for analysis of classification and adulteration of honey," *IEEE Transactions on Agrifood Electronics*, pp. 1–8, 2026. [DOI: 10.1109/TAFE.2026.3666205](#).
- 5 A. Paul, D. Adak, S. Dutta, and S. Majumdar, "Non-invasive rcc beam assessment via sdr-integrated microstrip filter," *IEEE Sensors Letters*, 2025. [DOI: 10.1109/LSENS.2025.3607598](#).
- 6 A. Paul, R. Snaitang, P. K. Gautam, and S. Majumdar, "Econoscan: Affordable microwave imaging for non-destructive strength testing of bituminous materials," *IEEE Sensors Letters*, 2025. [DOI: 10.1109/LSENS.2025.3613338](#).
- 7 A. Paul, G. Bhargava, and S. Majumdar, "Impact of holes in an interdigital parallel folded coupled line-based wideband bandpass filter for x-band applications," *Microwave and Optical Technology Letters*, vol. 66, no. 1, e33903, 2024. [DOI: 10.1002/mop.33903](#).
- 8 A. Paul, G. Bhargava, and S. Majumdar, "Development and characterization of log periodic feedline based filter for water leakage detection in size invariant pvc pipes," *Flow Measurement and Instrumentation*, vol. 98, p. 102647, 2024, ISSN: 0955-5986. [DOI: 10.1016/j.flowmeasinst.2024.102647](#).
- 9 A. Paul, H. Kumari, R. Snaitang, P. K. Gautam, and S. Majumdar, "Microwave imaging and non-destructive testing of bituminous mix binder-aggregate behavior using log-periodic feedline-based microstrip filter," *NDT*, vol. 2, no. 3, pp. 347–362, 2024. [DOI: 10.3390/ndt2030021](#).
- 10 A. Paul, J. Borah, H. Kumari, and S. Majumdar, "Surrogate-assisted framework for log-periodic feedlines based filter optimization," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (Accepted)*.
- 11 H. Kumari, A. Paul, S. Majumdar, G. Crupi, and J. Cai, "A compact dual-band power amplifier using spoof surface plasmon polaritons for broadband rf applications," *AEU - International Journal of Electronics and Communications*, vol. 204, p. 156116, 2026, ISSN: 1434-8411. [DOI: 10.1016/j.aeue.2025.156116](#).
- 12 H. Kumari, A. Paul, W. Sangma, and S. Majumdar, "Exhibiting the superiority of metal-filled sspp in advanced output matching networks for high-performance power amplifier," *AEU - International Journal of Electronics and Communications*, vol. 201, p. 156016, 2025, ISSN: 1434-8411. [DOI: 10.1016/j.aeue.2025.156016](#).
- 13 H. Kumari, A. Paul, and S. Majumdar, "Machine learning based design of ultra-wideband gan hemt power amplifiers for next-generation 6g systems," *Microsystem Technologies*, pp. 1–13, 2025. [DOI: 10.1007/s00542-025-05939-5](#).
- 14 H. Kumari, A. Paul, G. Bhargava, and S. Majumdar, "Apd-sspp: Enhancing linearization of power amplifier using spoof surface plasmon polariton propagation," *Microwave and Optical Technology Letters*, vol. 67, no. 10, e70409, 2025. [DOI: 10.1002/mop.70409](#).

Conference Proceedings

- 1 A. Paul, R. Snaitang, P. K. Gautam, and S. Majumdar, "Assessment of binder aggregate behavior of bituminous mixes using log-periodic feedlines based microstrip filter," in *2024 IEEE Microwaves, Antennas, and Propagation Conference (MAPCON)*, IEEE, 2024, pp. 1–4.
- 2 A. Paul, H. Kumari, J. Borah, and S. Majumdar, "Data-driven ensemble prediction models for design of microwave filter for sensing applications," in *2024 IEEE 21st India Council International Conference (INDICON)*, IEEE, 2024, pp. 1–6.
- 3 A. Paul and P. Sarkar, "Compact multi-band bandpass filters using substrate integrated waveguide and fractal resonators," in *Proceedings of International Conference on Data, Electronics and Computing*, Springer Nature Singapore, 2023, pp. 333–340.

- 4 H. Kumari, A. Paul, and S. Majumdar, "Digital predistortion behavioral modeling of power amplifiers: A neural network-based approach to nonlinearity compensation," in *2024 IEEE International Conference of Electron Devices Society Kolkata Chapter (EDKCON)*, IEEE, 2024, pp. 01–06.
- 5 H. Kumari, A. Paul, J. Borah, and S. Majumdar, "Data-driven bilstm prediction model for dual-band rf power amplifier," in *2024 IEEE Microwaves, Antennas, and Propagation Conference (MAPCON)*, IEEE, 2024, pp. 1–4.
- 6 H. Kumari, A. Paul, G. Bhargava, and S. Majumdar, "Advanced design strategies for ultrawideband power amplifiers utilizing multi-branch matching networks," in *2024 IEEE Space, Aerospace and Defence Conference (SPACE)*, IEEE, 2024, pp. 721–724.

Technical Skills

Electromagnetics & RF Design	Expertise in microwave engineering, RF circuit design, and electromagnetic theory, with hands-on experience in passive/active RF components, and circuit design.
EM Simulation Tools	Proficient in CST Microwave Studio, Ansys HFSS, and Keysight ADS.
Antenna & Filter Design	Experience in designing broadband antennas, reconfigurable structures, microstrip ring resonators, log-periodic structures, and optimized microwave filters for sensing and communication.
Microwave Sensing & Imaging	Skilled in VNA-based measurements, and 3D microwave imaging for structural health monitoring and dielectric property extraction.
Thermal & Reliability Analysis	Experience with junction temperature prediction, thermal optimization of RF PAs, and reliability estimation.
AI/ML for EM Design	Knowledge of surrogate modelling, EM optimization using ensemble learning, regression models, and data-driven prediction of RF behaviour under varying conditions.
Experimental Prototyping	Strong hands-on expertise in PCB fabrication, substrate machining, precision sample preparation, temperature-controlled measurement protocols, and maintaining experimental consistency.
Programming	Python and MATLAB for EM data analysis, automation of VNA measurements, 3D reconstruction, and machine learning pipelines for RF applications.

Awards and Achievements

- 2025
 - Best Paper Award** in Industry, Academia and Research Conclave (IARC), 2025 conducted in National Institute of Technology Meghalaya, India.
 - Raj Mitra Grant in India (RMGI)** for IEEE MAPCON 2025 held in Kochi, India (14-18 December 2025).
- 2022
 - Gold Medalist** in M.Tech in ECE from North Eastern Hill University, Shillong, India.

Professional Affiliations

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Web of Science Researcher ID	KLY-4285-2024
Google Scholar ID	https://scholar.google.com/citations?user=tFqGI8kAAAAJ&hl=en&authuser=1

References

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